

SHETH LUJ COLLEGE

AI-Practical-6

Adaboost Ensemble Learning

- ❑ Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- ❑ Train the ensemble model on a given dataset and evaluate its performance.
- ❑ Compare the results with individual weak classifiers.

```
IDLE Shell 3.13.7*
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Daniyal khan/AI_PRACTICAL/Adaboost_AI.py =====
Libraries imported
Dataset Loaded

First 5 rows:
  Date Item_ID ... Seasonal_Factor Waste_Percentage
0 2025-06-10 1 ... 1.11 2.98
1 2025-06-10 2 ... 0.81 3.54
2 2025-06-10 3 ... 1.23 4.96
3 2025-06-10 4 ... 0.90 3.06
4 2025-06-10 5 ... 1.07 3.09

[5 rows x 14 columns]

Dataset Shape:
(1000, 14)

Data after preprocessing:
  Current_Stock Reorder_Level ... Seasonal_Factor Waste_Percentage
0 21.45 8.12 ... 1.11 2.98
1 12.84 5.34 ... 0.81 3.54
2 22.35 4.49 ... 1.23 4.96
3 8.36 3.16 ... 0.90 3.06
4 12.31 6.19 ... 1.07 3.09

[5 rows x 7 columns]
```

```
IDLE Shell 3.13.7*
Dataset Shape after preprocessing:
(1000, 7)

Reorder Status:
Reorder_Needed
0 968
1 32
Name: count, dtype: int64

Features:
  Current_Stock Reorder_Level ... Seasonal_Factor Waste_Percentage
0 21.45 8.12 ... 1.11 2.98
1 12.84 5.34 ... 0.81 3.54
2 22.35 4.49 ... 1.23 4.96
3 8.36 3.16 ... 0.90 3.06
4 12.31 6.19 ... 1.07 3.09

[5 rows x 7 columns]

Target:
0 0
1 0
2 0
3 0
4 0
Name: Reorder_Needed, dtype: int64

Training Data Size:
(800, 7)
```



```

4      12.31      6.19 ...      1.07      3.09

[5 rows x 7 columns]

Target:
0  0
1  0
2  0
3  0
4  0
Name: Reorder_Needed, dtype: int64

Training Data Size:
(800, 7)

Testing Data Size:
(200, 7)

Weak Classifier Accuracy:
97.0 %

Weak Classifier Confusion Matrix:
[[190  4]
 [ 2  4]]

Weak Classifier Classification Report:
      precision    recall  f1-score   support

0       0.99      0.98      0.98       194
1       0.50      0.67      0.57         6

```

[illegible]

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[illegible]

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